Stage C: where we are and what happens next

Written 9 November 2026, 13:40, for Gabriel and Léonard. Usable as the Check-in 2 status page.

1. The short version

Pretraining is nearly finished and far ahead of schedule. Evaluation has not started, and it is now the only thing on the critical path. Thirteen of the fourteen Stage C checkpoints exist; none of them has yet produced a number that can be compared to anything published, and none can until something attaches a task head and trains it on labelled data.

track	plan said	actual	status
Pretraining, 500 to 50k	submitted Wed 12 to Fri 14 Nov	all twelve done by 12:24 on Sun 9 Nov	about five days ahead
Pretraining, 100k	decide Mon 17, land Tue 18 to Fri 21	seed 1 done, seed 0 at epoch 262	about nine days ahead
Fine-tuning loop (B2)	start Mon 10 Nov, first numbers Wed 12	not started	on schedule, not late
Baselines, scratch and released	Thu 13 Nov	not started	blocked on B2
Curve figure	Mon 17 Nov, Check-in 2	not started	blocked on B2

About 65 GPU-hours consumed so far, against the 61 to 102 projected for the whole curve. The budget is holding.

2. Is the pretraining done?

Not quite. Twelve of fourteen conditions are complete and exported. The two 100,000-molecule runs are still training.

size	seed 0	seed 1
500	done, epoch 227	done, epoch 82
1,000	done, epoch 241	done, epoch 164
5,000	done, epoch 308	done, epoch 337
10,000	done, epoch 262	done, epoch 289
25,000	done, epoch 214	done, epoch 275
50,000	done, epoch 283	done, epoch 246
100,000	running, at epoch 262	done, epoch 264

Each finished run stopped 50 epochs after its best validation epoch, exported a 48 MB `best.pt`, and none reached the 500-epoch cap.

Seed 1 finished at epoch 264 after 19 hours 11 minutes. Seed 0 is at epoch 262 on its third attempt, having been preempted twice, and should finish tonight on the same trajectory. The resume wrapper has now recovered three real interruptions without losing a run.

The resume wrapper earned its keep

Two real interruptions, both recovered automatically and neither costing a run:

- **A wall kill.** The 25,000-molecule seed 1 run hit `mit_normal_gpu`'s

six-hour limit at epoch 256. The next job in its chain resumed from `last.ckpt`, started at epoch 256, and finished the condition 27 minutes later at epoch 275.

- **A preemption.** The 100,000-molecule seed 0 run was preempted overnight,

requeued, and restarted at 02:43 from its checkpoint rather than from zero. It is now at epoch 176 with about 11 hours of GPU work preserved.

One flaw found in how the 25k jobs were chained, worth fixing before the pattern is reused. Each link was

submitted with `afternotok` on the previous one, so it would only run if that one failed. When the first link *succeeded*, the second was correctly killed for an invalid dependency - but Slurm counts a killed job as "not ok", so the third link then ran anyway. It resumed a condition that had already converged, re-ran the test, and exited after two minutes. No harm: its test MSE was bit-for-bit identical to the original (0.08873146027326584 both times), confirming it reloaded the same best checkpoint. But it burned a GPU slot for nothing. `--requeue` on a preemptable job does this job properly and needs no chain.

3. What the pretrained models can and cannot tell us yet

This is the part worth reading carefully, because it is the easiest place in the project to draw a wrong conclusion.

What we have

Each condition reports a test MSE on standardised Mordred descriptors, over its own held-out tenth:

size	seed 0	seed 1	spread
500	0.296	0.448	0.152
1,000	0.310	0.337	0.027
5,000	0.149	0.151	0.002
10,000	0.101	0.104	0.003
25,000	0.089	0.079	0.010
50,000	0.055	0.062	0.007
100,000	running	0.043	

Two things in it are worth noting, and one thing is not a result at all.

The objective is still improving at 100,000 molecules. Error falls by a factor of seven from 500 to 100,000 with no sign of flattening. **And the seed spread collapses**, from 0.152 at n=500 to a few thousandths above n=5,000: below a few thousand molecules, which molecules you happen to draw matters more than how many.

This is not the size curve. It is a health check, for three reasons. The test sets differ between conditions, so the columns are not strictly comparable. The quantity is descriptor reconstruction error, which is the pretraining objective, not a chemical property anyone wants predicted. And at n=500 the test set is 50 molecules, which is why its two seeds sit 0.15 apart.

Most importantly, learning the pretraining objective better says nothing on its own about whether that transfers. The whole question of the project is whether the corpus changes downstream performance, and descriptor MSE cannot answer it.

Why there is no comparison to Chemprop yet

The lecture slides and the proposal report AUROC on BBBP and BACE and RMSE on ESOL. Those are fine-tuned, downstream, property-prediction numbers. Setting a descriptor MSE beside a published AUROC would be a category error: different quantity, different data, different units.

Producing a comparable number requires the fine-tuning loop. The protocol for it is already frozen in `docs/protocol.md` - at most 50 epochs, early stopping with patience 10, batch 64, Adam at 1e-4, no per-condition tuning, five scaffold-split seeds, label counts of 50, 100, 500 and all - and `scripts/evaluate.py` is already written and waiting for the `predictions.csv` files it will produce. The loop itself is `scripts/finetune.py`, 36 lines ending in `raise NotImplementedError`. It is blocker B2, assigned to Léonard, estimated two days, and due to begin Monday 10 November.

So the honest position is: twelve encoders are trained and sitting on disk, and the y-axis of the plot does not exist yet.

4. Where to look for an effect, once we can look

Section 6 of the Check-in 1 report read the statistics off the 1 October lecture, and they set expectations that matter for how we spend the remaining days.

- **Six methods, including Chemprop from scratch, could not be separated at

all** on ten MoleculeNet datasets under the harder scaffold split: the Friedman test gives p = 0.194.

- **The split dominates the method.** The deck's own illustration moves AUROC

from 0.78 to 0.54 between a random and a scaffold split - 0.24 points from the protocol alone, an order of magnitude more than any corpus effect we could plausibly find.

- **Gains live in the low-label regime.** On BACE the best foundation model led

Chemprop by about 0.07 AUROC at 60 training molecules and 0.04 at 847. On ESOL at 45 molecules, three pretrained models were *worse* than training from scratch.

- **Adding computed descriptors to Chemprop is worth as much as pretraining

is*: *about 0.09 AUROC on BACE at n=60. CheMeleon's pretraining target* is* computed descriptors, so without a descriptors-only baseline a corpus effect and a descriptor effect cannot be told apart.

Three consequences for what we run:

1. **The low-label curves are where a signal can exist.** BBBP at 1,631 training molecules and ESOL at 902 are both near the top of the range where the effect is reported to be smallest. The 50- and 100-label columns matter more than more seeds at full labels.
2. **The descriptor baseline is not optional.** It is the one baseline that can attribute our own effect, and it is cheap.
3. **A null result is a real result** and the protocol already says to report it as one. Given p = 0.194 across six methods in the published work, a flat curve is a plausible honest outcome, not a failure.

5. Next steps, in order

Blocking everything

#	task	owner	estimate
6	Fine-tuning loop (B2). Write scripts/finetune.py against the frozen protocol. Verify on the released checkpoint and from scratch before touching ours.	L	2 days, due to start Mon 10 Nov

Immediately unblocked once task 6 lands

#	task	owner	note
7	Scratch Chemprop baseline, both tasks, five split seeds	L	the horizontal line the whole curve is read against
8	Released 1M checkpoint fine-tuned, both tasks, five seeds	L	the top point of the curve
8b	Chemprop plus Mordred descriptors baseline, both tasks, five seeds	L	not in the original task list; section 4 argues it is the one that makes our effect attributable
13	Fine-tune all fourteen checkpoints, both tasks, five seeds, four label counts	L	twelve are ready now

Pretraining side, remaining

#	task	owner	note
12	100k both seeds	G	running, expected to finish 9 or 10 Nov
14	Sensitivity run: largest completed size with --exclude-benchmarks	G	belongs at 100k, where benchmark exposure is largest, not where it is cheapest
15	Curve figure, paired differences with intervals, operating size for Stage D	both	needs task 13

Worth doing while we wait

- **Fix the chain pattern** so a successful first link cannot trigger a third

(section 2). Use `--requeue` instead.

- **Prepare the comparison table** with the published lecture figures filled in

and our cells empty, so numbers drop into a reviewed structure rather than being assembled under time pressure.

- **Keep the 25k and 50k checkpoints as the Stage D candidates.** The

descriptor objective is still improving at 50k, so the operating size for Stage D cannot be read off section 3 - it needs the downstream curve.

6. One deviation from the plan, recorded

The plan gated 100k behind G3 on 17 November ("decide whether to start 100k at all"). It started on 8 November instead. The justification is G1's own pre-agreed rule: measured throughput put a 100k run at 22 to 45 GPU-hours a seed, which lands in G1's 30-to-60 band, whose written action is "proceed, but 100k runs on preemptable only and starts the same day as 25k, not after it". That is what was done. G3's decision point is therefore superseded rather than skipped, but it is a deviation and should be confirmed rather than assumed.

7. What to do with the pretrained models now

Thirteen encoders are on disk. They are inert: each one ends in a 1,613-wide descriptor head, so none can emit a BBBP or ESOL prediction. Something has to attach a task head and train it on labelled data. There are two ways to do that, and they differ in cost by roughly a factor of fifty.

Route A - a linear probe, available today

Freeze each encoder, push BBBP's 2,050 and ESOL's 1,128 molecules through it, keep the 2,048-wide vector each produces, and fit logistic regression for BBBP and ridge for ESOL on the frozen scaffold splits already on disk.

- No GPU training: about 38,000 forward passes, then scikit-learn.
- An hour of work, minutes of compute.
- Produces a real AUROC and RMSE per condition, so the curve can be inspected

for slope tonight rather than next week.

What it is not. Not comparable to the lecture's figures or to CheMeleon's paper, both of which fine-tune. Probes also tend to *understate* the pretraining benefit, because fine-tuning gets to adapt the encoder and a probe does not. So it is an addition to the frozen protocol, labelled as such in any figure it appears in, and not the headline result.

What it is for. Information about the project rather than a result for the write-up. If the probe curve is flat from 500 to 100,000 molecules, we learn that before anyone spends two days on the loop, and the pivot discussion starts a week earlier. If it rises, we know the signal exists and roughly where it turns, which is exactly what

Stage D needs in order to pick its operating size.

A probe is also a reasonable test of the *low-label* columns specifically. Where the bottleneck is labelled data rather than model capacity - 50 or 100 molecules - a frozen representation plus a linear fit is close to what a fine-tuned model can do anyway, so the probe's understatement is smallest exactly where the published effect is largest.

Route B - the fine-tuning loop, the actual deliverable

`scripts/finetune.py`, written against the protocol already frozen in `docs/protocol.md`: at most 50 epochs, early stopping at patience 10, batch 64, Adam at 1e-4, no per-condition tuning, five scaffold seeds, label counts of 50, 100, 500 and all. Chemprop's `--from-foundation` flag does the mechanical part, taking the encoder and discarding the descriptor head.

This is blocker B2 - Léonard, two days, due to begin 10 November. It unlocks Stages B, C, D and E. Nothing in the write-up can come from Route A alone.

Recommendation

Run the probe now and let the loop produce the numbers that go in the write-up. The probe costs almost nothing and it answers a question worth answering before the expensive work is sequenced: is there a slope at all? The two are not in competition - the probe reads the existing checkpoints, the loop is a separate script, and neither blocks the other.

If the probe is run, it goes in its own script with its own output directory, so it cannot be mistaken for the protocol's fine-tuning path.

8. The thing to decide now

Two questions need answers, and only the second is urgent:

1. **Should the probe run tonight?** It is cheap, reversible, and the only way to see the curve's shape before the end of the week.
2. **Is Léonard starting the loop tomorrow as planned?** If yes, nothing changes and the schedule holds with nine days of slack on the pretraining side. If he is blocked, decide tonight who writes it. Waiting is the expensive option: every day it slips, the curve, the baselines and the Check-in 2 figure all slip together, since all three are strictly downstream of that one file.

Two things that need no GPU and are not on anyone's list:

- **Pin both baselines in `docs/protocol.md`** under Amendments, which

currently reads "None yet". The scratch baseline's architecture is unspecified, and the curve floats against an arbitrary line until it is fixed. Needed before task 7 runs, not after.

- **Add the Chemprop-plus-descriptors baseline** (task 8b). Adding computed

descriptors to Chemprop is worth about as much as pretraining is, and since CheMeleon's pretraining target *is* computed descriptors, without it a corpus effect and a descriptor effect cannot be told apart.